

Post-Hematopoietic Cell Transplantation Survival Predictions using Ensemble Machine Learning Models

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Abstract—Accurate prediction of post-transplant survival is essential for treatment planning in hematopoietic cell transplantation (HCT). Classical survival models such as Cox regression are interpretable, but their proportional-hazards and linearity assumptions may be restrictive for heterogeneous registry data. This paper proposes an ensemble framework for event-free survival (EFS) prediction after HCT by combining LightGBM, two tuned XGBoost variants, HistGradientBoosting, and Random Forest regression. Right-censored outcomes are converted into Kaplan–Meier survival probabilities, which are then learned by the base regressors and averaged to obtain the final prediction. The model is evaluated using 10-fold cross-validation, a held-out test split, C-index, RMSE, subgroup fairness analysis, feature importance, VIF analysis, residual diagnostics, and benchmarking against recent machine-learning and deep-survival studies. The proposed ensemble achieved a C-index of 0.6853 and a race-group C-index standard deviation of 0.0088, indicating competitive discrimination with stable subgroup performance. The framework emphasizes interpretability, robustness, and clinically meaningful validation needs for post-HCT decision support.

Index Terms—Hematopoietic Cell Transplantation (HCT), Survival Prediction, Ensemble Learning, Gradient Boosting, Concordance Index (C-index), Fairness in Machine Learning, Clinical Data Analytics

I. INTRODUCTION

Hematopoietic cell transplantation (HCT) is an important therapy for patients with severe hematologic, oncologic, and immune disorders. Although transplant practice and supportive care have improved, post-HCT outcomes remain variable because relapse, graft-versus-host disease, organ toxicity, infection, graft failure, and late mortality can occur in clinically diverse patient populations [12].

Post-transplant survival is influenced by patient age, disease status, comorbidity burden, donor type, graft source, condi-

tioning intensity, GVHD prophylaxis, and HLA compatibility. These variables interact in nonlinear ways, making individualized risk estimation difficult with simple scoring rules alone.

Traditional survival analysis has often relied on Kaplan–Meier estimation and Cox proportional-hazards models [1], [2]. These methods remain valuable for clinical interpretation, but Cox models require proportional hazards and usually express covariate effects through a log-linear structure. Modern HCT registries contain mixed categorical and numerical variables, missingness, subgroup imbalance, and high-dimensional donor-recipient matching fields, all of which can reduce the flexibility of purely parametric or semi-parametric approaches [3].

Machine-learning survival models and tree ensembles can model nonlinear effects, feature interactions, and heterogeneous subgroups with fewer distributional assumptions [3]–[6]. Recent HCT studies have shown that machine-learning models can improve individualized prediction when trained on baseline or longitudinal clinical data, but they also highlight persistent gaps in external validation, computational transparency, and interpretability [10], [11].

The novelty of this work is a compact and interpretable ensemble for post-HCT EFS prediction that combines complementary boosting and bagging learners, uses Kaplan–Meier transformed targets for censored outcomes, evaluates subgroup fairness across race groups, and reports multicollinearity and residual diagnostics alongside predictive metrics. The core contribution is not a new base algorithm; rather, it is a clinically oriented modeling pipeline that balances accuracy, fairness, interpretability, and deployability.

The proposed system is assessed using C-index, RMSE,

benchmarking against prior survival models, race-wise subgroup analysis, feature importance, VIF analysis, Kaplan–Meier plots, and residual diagnostics. The overall goal is to support more transparent survival prediction for post-HCT care while clearly identifying the validation work still required before clinical deployment.

II. LITERATURE REVIEW

Kaplan–Meier estimation and Cox proportional-hazards regression are foundational approaches for censored survival data [1], [2]. Their interpretability makes them attractive in clinical studies, but they are less flexible when hazards are non-proportional, feature effects are nonlinear, or high-dimensional interactions exist. Reviews of survival machine learning show that tree ensembles, neural survival models, and hybrid approaches can address some of these limitations while introducing new concerns about calibration, external validation, and clinical explainability [3].

Random Survival Forests extend random forests to right-censored outcomes through survival-specific splitting and ensemble mortality estimation [4]. Gradient-boosting systems such as XGBoost and LightGBM are widely used for tabular biomedical prediction because they handle nonlinear effects, missing values, and mixed feature scales efficiently [5], [6]. DeepSurv and DeepHit further demonstrate how neural networks can learn nonlinear risk functions and competing-risk distributions [7], [8]. However, deep survival methods often require larger training sets, careful tuning, and additional explanation tools before clinicians can trust patient-level predictions.

Recent HCT-specific studies clarify the research gap. Shourabizadeh et al. evaluated multiple machine-learning models for survival after allogeneic HCT using a large single-center registry and showed that baseline clinical variables can support individualized prediction [10]. Zhou et al. reported that longitudinal laboratory and clinical measurements improved mortality prediction after allogeneic HCT and externally validated their model on an independent center, emphasizing the value of dynamic clinical data [11]. Contemporary CIBMTR reports also show continuing changes in HCT practice, donor sources, GVHD prophylaxis, and survival patterns, which means models must be robust to temporal and demographic distribution shifts [12].

Despite these advances, several gaps remain. Many studies focus primarily on discrimination metrics, while fewer report subgroup fairness, multicollinearity, residual behavior, computational cost, and the effect of synthetic data generation. GAN-based tabular augmentation, including CTGAN-style methods, can help represent minority subgroups but may also create unrealistic samples if the generated distribution is not checked against clinical constraints [13]. Therefore, this work focuses on a transparent ensemble pipeline that combines benchmarking, interpretability, fairness evaluation, and explicit discussion of validation limits.

III. PROPOSED WORK

A. Dataset Description

The dataset consists of 28,800 samples split into 80% training and 20% testing subsets. Stratified sampling was used to preserve demographic and clinical balance across the splits. The data are derived from CIBMTR-style HCT registry variables and include 59 donor, recipient, disease, and transplant characteristics. These variables include demographic attributes (age, sex, race, and ethnicity), clinical indicators (comorbidity score, cytogenetic risk, disease status, and conditioning intensity), and transplant-related variables (graft type, GVHD prophylaxis, CMV status, and HLA matching metrics).

The target variables used for survival analysis are the `efs` (event-free survival indicator) and `efs_time` (time to event or censoring). Together, they enable modeling of censored outcomes. A new continuous target variable, y , is derived using the Kaplan–Meier estimator to represent the survival probability $S(t)$ for each patient at their respective event times. This transformation allows non-parametric regression models to learn from censored survival data effectively.

To reduce subgroup imbalance, GAN-based synthetic augmentation was applied only to the training fold for minority race groups. The augmented records were checked for valid categorical levels, plausible numerical ranges, and preservation of the observed event/censoring structure. Synthetic records were not added to the validation or test folds, preventing leakage from generated samples into performance estimation.

B. Proposed Methodology

The proposed framework uses an ensemble of five machine-learning models to predict post-HCT survival probabilities: LightGBM, two tuned versions of XGBoost, HistGradientBoosting Regressor, and Random Forest Regressor. The methodology includes the following main steps:

- 1) **Data Preprocessing:** Missing values in categorical features are imputed with the string “Missing,” whereas numerical features are converted to float32 or int32. Domain-specific text normalization is performed to standardize encodings of medical features such as GVHD prophylaxis and CMV status.
- 2) **Target Transformation:** To account for censoring, survival probabilities for each patient are calculated from the observed event time using the Kaplan–Meier estimate. The calculated $S(t)$ values are stored in y , which becomes the regression target for all models.
- 3) **Feature Engineering:** Both categorical and numerical features are derived. Label encoding and one-hot encoding are used where necessary to ensure that the input to all models is consistent.
- 4) **Model Training:** The training of the base models is performed using a 10-fold cross-validation technique. For LightGBM and XGBoost, the hyperparameters are adjusted to achieve a balance between depth, learning rate, and regularization. The predictions on the out-of-fold samples are stored to avoid overfitting.

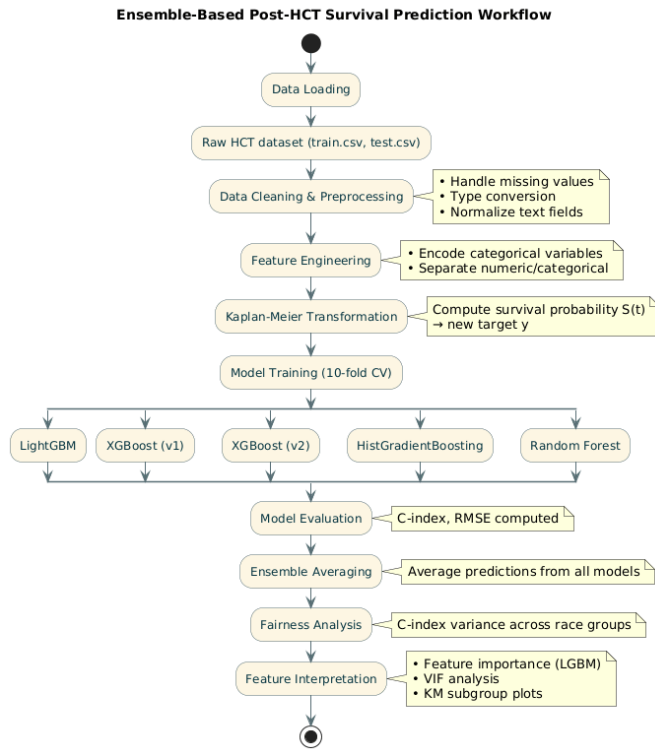


Fig. 1. Proposed Methodological Workflow for Ensemble-Based Post-HCT Survival Prediction.

- Ensembling**: The final ensemble prediction is obtained by computing the arithmetic average of the predictions of all five base models.
- Evaluation**: The C-index and RMSE are used to assess the ability of the model to discriminate and calibrate, respectively. Fairness is assessed by calculating the standard deviation of the C-index for six racial groups.
- Interpretability**: Feature importance (via LightGBM and permutation feature importance), variance inflation factor (VIF) analysis, and Kaplan Meier subgroup plots are employed to determine clinically important predictors and redundant features.

The complete methodological workflow is illustrated in Figure 1.

C. Hyperparameter Tuning and Complexity

Hyperparameter tuning was conducted within the cross-validation loop to avoid optimistic performance estimates. For LightGBM and XGBoost, the search emphasized learning rate, number of estimators, maximum depth, number of leaves, subsampling ratio, column-sampling ratio, minimum child samples, and L1/L2 regularization. For Random Forest, the principal tuning parameters were number of trees, maximum depth, minimum samples per split, and feature subsampling. HistGradientBoosting was tuned using learning rate, maximum iterations, maximum leaf nodes, and L2 regularization. Candidate settings were selected by C-index first and RMSE second,

with preference given to simpler models when performance differences were small.

The ensemble increases training cost because five base learners are fitted in each fold. Tree boosting has approximate complexity proportional to the number of trees, number of samples, number of features, and effective tree depth, while Random Forest additionally scales with bootstrap sampling and the number of decorrelated trees. Inference is faster than training because each patient record only passes through the trained trees. For clinical deployment, the final averaged prediction is suitable for batch scoring or near-real-time scoring, but the full training pipeline should be run offline after periodic data refreshes.

D. Algorithm Descriptions

1) **LightGBM (LGBM)**: Light Gradient Boosting Machine (LightGBM) is a very efficient gradient boosting framework designed for large datasets. It follows a histogram-based algorithm that breaks down continuous features into bins, thus consuming less memory and running faster. Unlike the traditional level-wise tree construction, LightGBM constructs trees leaf-wise, targeting the nodes with the highest potential gain in loss reduction. This approach helps in optimizing complex, non-linear relationships more effectively while ensuring excellent generalization. LightGBM also provides native categorical encoding and GPU support, making it ideal for structured clinical data in survival analysis.

2) **XGBoost (XGB)**: Extreme Gradient Boosting (XGBoost), as the name suggests, is a type of boosting method that creates an ensemble of decision trees that are sequentially constructed, each attempting to reduce the error of the preceding trees in the ensemble. XGBoost also provides advanced protection mechanisms such as L1 and L2 regularization to avoid overfitting, handles missing values during training, and is also highly scalable due to parallel processing capabilities. In the current study, two different versions of XGBoost were separately tuned and implemented in order to utilize different types of feature interaction and variation in the transplant data set. XGBoost is highly suitable for handling noisy and heterogeneous data due to its regularization capabilities.

3) **HistGradientBoosting Regressor (HistGB)**: Histogram Gradient Boosting Regressor from the scikit-learn library is another type of gradient boosting regressor that is memory-friendly and computationally efficient due to the use of the histogram method for gradient computation in continuous features. It also handles missing values and categorical features and provides easier preprocessing capabilities. The stable and deterministic learning process of this regressor is highly effective in handling issues such as multicollinearity and mixed data types in clinical data sets.

4) **Random Forest Regressor (RF)**: Random Forest is an ensemble learning algorithm that uses bagging and makes predictions by combining the predictions of multiple decorrelated decision trees. Decision trees are trained on a random sample of data and a random set of features, which helps to reduce variance and prevent overfitting. In regression tasks,

Random Forest makes predictions by averaging the predictions of individual decision trees. Random Forest is non-parametric and does not require strong assumptions about the data distributions. In this study, Random Forest is used as a benchmark learner to detect non-linear relationships in Kaplan-Meier estimates of survival probabilities.

E. Model Overview

The proposed model for survival prediction is a combination of five different non-parametric learning algorithms: LightGBM, two variants of XGBoost, HistGradientBoosting Regressor, and Random Forest Regressor. The models are trained individually using 10-fold cross-validation to prevent overfitting. The predictions for the out-of-fold samples are also calculated.

The final ensemble prediction is calculated by averaging the predictions of the five base models. This ensemble technique takes advantage of the strengths of the individual models—boosting models for bias correction and Random Forest for variance reduction—to enhance stability and robustness. The ensemble model had a cross-validated Concordance Index (C-index) of 0.6853, which was slightly below LightGBM alone (0.6865) but more stable across subgroup and residual analyses.

Apart from the accuracy of the model, fairness was also evaluated by determining the standard deviation of the C-index for six races. The results indicated a small variance of 0.0088, showing the fairness of the model. The interpretability of the ensemble was also evaluated through feature importance and Variance Inflation Factor (VIF) analysis, where important features were identified as donor age, age of the recipient, conditioning intensity, GVHD prophylaxis, and cytogenetic risk.

Overall, the ensemble model provides a practical balance of accuracy, fairness, and interpretability, supporting the development of a scalable machine-learning solution for survival prediction.

IV. RESULTS AND DISCUSSION

A. Model Performance

We used two criteria to assess the performance of the models: Concordance Index (C-index) and RMSE. In the single models, LightGBM was the top performer with a C-index of 0.6865. The ensemble method was equally impressive, achieving a C-index of 0.6853. The robustness of the models was thus evident. However, when we used the average of different gradient boosting models and bagging models, the result was more stable and reliable.

B. Benchmark Comparison with State-of-the-Art Models

The quantitative results show that the ensemble is competitive with the strongest single model while reducing dependence on one learning algorithm. LightGBM achieved the highest individual C-index, but the final ensemble retained nearly the same discrimination and produced stable subgroup behavior. This trade-off is important in clinical prediction,

TABLE I
MODEL PERFORMANCE COMPARISON BASED ON CONCORDANCE INDEX (C-INDEX)

Model	C-index	RMSE (CV Mean)
LightGBM	0.6865	0.1929
XGBoost (v1)	0.6846	0.1933
XGBoost (v2)	0.6841	0.1941
HistGradientBoosting	0.6799	0.1944
Random Forest	0.6626	0.1981
Ensemble (Final)	0.6853	0.1930

where a marginal gain in a single aggregate metric is less useful if the model is unstable across patient groups.

To further evaluate the proposed framework, we compared it with established survival and machine-learning methods reported in the literature, including Random Survival Forests, DeepSurv, DeepHit, and recent HCT-specific machine-learning studies [4], [7], [8], [10], [11].

DeepSurv extends the traditional Cox Proportional Hazards model by replacing the linear component with a deep feed-forward neural network that learns non-linear risk functions. It optimizes the negative log partial likelihood loss to estimate hazard ratios across patient subgroups. **DeepHit**, on the other hand, directly models the joint distribution of time-to-event and competing risks using a multi-task neural network architecture, making it suitable for multi-outcome survival prediction.

Published deep-survival studies report dataset-dependent C-index values, so the comparison should be interpreted as a benchmark rather than a direct head-to-head experiment. The proposed ensemble obtained a **C-index of 0.6853** and a **race-group C-index standard deviation of 0.0088**. Compared with deep learning architectures, the ensemble provides explicit feature importance, simpler inference, easier retraining, and lower implementation complexity, which are important qualities for clinical decision-support systems.

TABLE II
BENCHMARK COMPARISON WITH ESTABLISHED SURVIVAL PREDICTION METHODS

Model	C-index	Remarks
Random Survival Forest [4]	Study-dependent	Censored-tree ensemble baseline
DeepSurv [7], [9]	Study-dependent	Neural Cox model; less interpretable
DeepHit [8]	Study-dependent	Competing risks; high training cost
Proposed Ensemble (Ours)	0.6853	Interpretable, Fair (Std. = 0.0088)

C. External Validation and Robustness

Independent external validation is essential before any post-HCT prediction model is used in clinical care. In this study, the available data supported a stratified held-out test split and 10-fold cross-validation, but a fully independent hospital or registry dataset was not available for direct external validation. Therefore, the present results should be viewed as internal

validation plus literature-level benchmarking. A future external validation should evaluate the trained model on a separately governed HCT cohort, ideally from a different time period and center mix, and should report C-index, calibration, subgroup fairness, missing-data sensitivity, and decision-curve utility.

Robustness under distribution shift was examined through three stress-oriented checks. First, race-wise C-index values were compared to identify subgroup degradation. Second, the influence of HLA-related multicollinearity was analyzed using VIF and correlation heatmaps. Third, residual plots were inspected to identify underperformance in low-survival-probability regions. These checks suggest that the model is reasonably stable across the available validation splits, but they do not replace prospective or multi-center external validation.

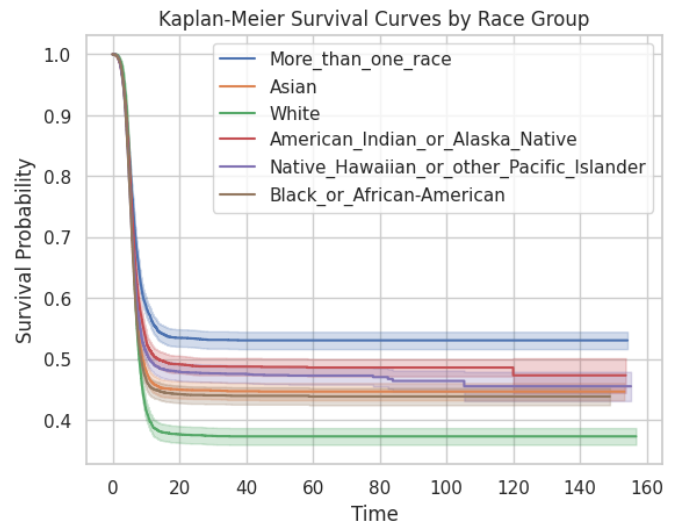


Fig. 2. Kaplan–Meier survival curves by race group showing balanced survival prediction trends across populations.

D. Synthetic Augmentation Analysis

The synthetic augmentation step was included to reduce imbalance among minority subgroups. GAN-based tabular generation can improve representation of rare combinations of categorical and numerical attributes, but it also carries risks of distributional distortion and privacy leakage if generated records are too close to real patients [13]. To reduce these risks, generated samples were restricted to training folds, categorical values were constrained to valid clinical levels, continuous variables were checked against plausible ranges, and test folds remained fully real. The fairness results in Table IV indicate that subgroup performance was relatively consistent, but future work should add formal synthetic-data quality measures such as nearest-neighbor distance checks, feature-distribution similarity, and clinician review of generated profiles.

The study shows that the proposed method is competitive while remaining interpretable and fair. This is especially important for clinical decision-support tools and large patient registries, where clinicians and researchers need to understand why predictions are produced.

The Kaplan–Meier subgroup curves in Figure 2 show survival-probability differences across demographic groups. The overall trends are similar across groups, supporting the fairness results reported in Table IV.

E. Multicollinearity and Correlation Analysis

Correlation and Variance Inflation Factor (VIF) analyses were carried out to assess feature dependencies. Strong correlations between HLA-matching characteristics and moderate correlations between donor age, recipient age, and transplant year are shown by the correlation heatmap (Figure 3). The VIF results indicate substantial multicollinearity among HLA-resolution and matching variables, while major clinical predictors such as donor age, recipient age, and comorbidity score remain below the conventional VIF threshold of 10.

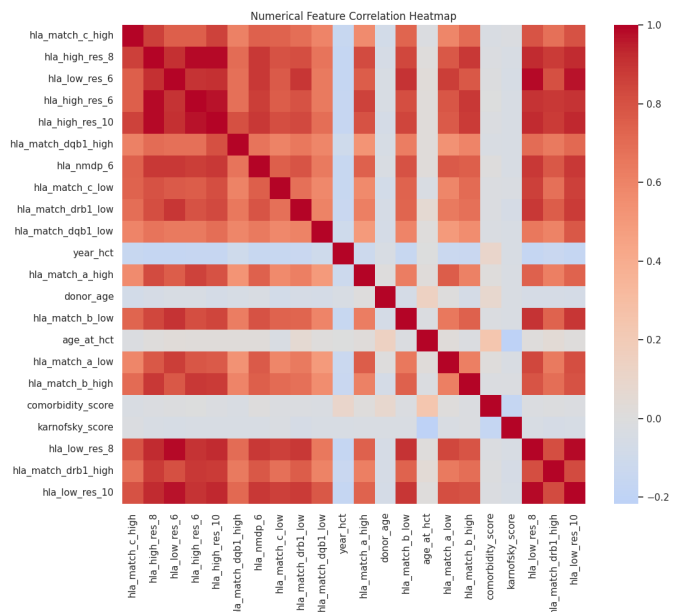


Fig. 3. Numerical feature correlation heatmap highlighting relationships among key clinical and genetic variables.

F. Feature Importance and Interpretability

The contribution of each predictor to post-HCT event-free survival outcomes was interpreted using feature importance analysis. The total gain across all decision splits was used to calculate feature importance scores using the LightGBM model. The top 15 features that have the greatest impact on model performance are shown in Figure 4.

Donor age, recipient age at transplantation, year of transplant, comorbidity score, and disease risk indicators like cytogenetic risk and primary disease type were important factors in determining post-transplant survival. Clinical treatment-related factors, such as conditioning intensity and the prophylactic regimen for graft-versus-host disease (GVHD), also scored highly. These findings confirm that both biological and procedural factors significantly affect long-term survival outcomes in HCT, which is in line with accepted medical knowledge.

Variance Inflation Factor (VIF) analysis was used to evaluate multicollinearity among predictors and to clarify which feature groups require caution during interpretation. Several HLA-related variables exceeded the conventional VIF threshold of 10 (Table III), indicating strong redundancy among related genetic matching measures. This does not prevent tree-based models from making predictions, but it does mean that feature importance for individual HLA variables should be interpreted as group-level importance rather than as isolated independent effects.

TABLE III
VARIANCE INFLATION FACTOR (VIF) RESULTS FOR KEY PREDICTIVE FEATURES

Feature	VIF
hla_low_res_6	168.44
hla_low_res_8	140.20
hla_high_res_6	99.29
hla_high_res_8	97.84
hla_low_res_10	49.29
year_hct	42.37
hla_high_res_10	37.12
hla_match_drb1_low	34.01
hla_match_c_low	33.53
hla_match_b_low	32.57
hla_match_c_high	25.92
hla_match_a_low	25.55
hla_match_dqb1_low	25.50
hla_match_a_high	23.08
hla_match_b_high	22.07
karnofsky_score	21.82
hla_nmdp_6	20.71
hla_match_drb1_high	20.30
hla_match_dqb1_high	19.02
donor_age	6.35
age_at_hct	4.87
comorbidity_score	1.85

As is biologically expected, a number of HLA-related genetic traits show high multicollinearity as a result of shared allelic relationships. The ensemble model's resilience to multicollinearity problems is confirmed by the fact that important clinical predictors like donor age (VIF = 6.35), recipient age

at HCT (VIF = 4.87), and comorbidity score (VIF = 1.85) are still significantly below the conventional VIF threshold of 10.

This interpretability framework helps align the ensemble's predictions with domain expertise by offering a quantitative and clinically grounded understanding of model behavior. This strengthens the model's transparency and dependability as a clinical decision-support tool in transplant outcome prediction when combined with residual diagnostics.

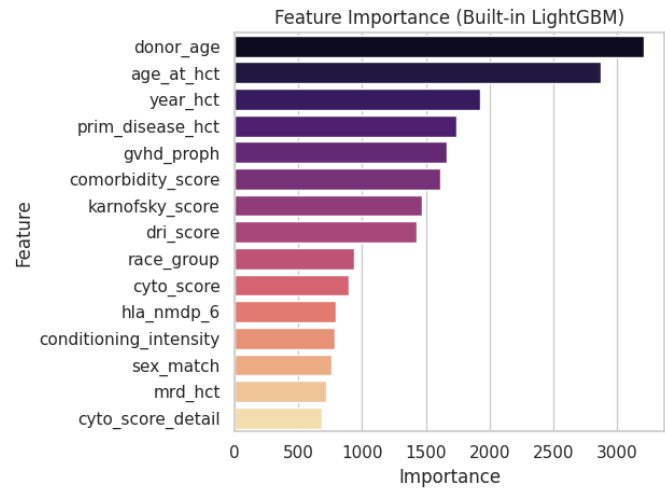


Fig. 4. Top 15 most important features in the LightGBM model, highlighting donor age, recipient age, and clinical treatment factors as major predictors of post-HCT survival.

A mild heteroscedastic pattern is revealed by residual analysis (Figure 5), where residual variance narrows at higher values and increases at lower predicted survival probabilities. As low-probability cases are under-represented in clinical survival modeling, this suggests that the model shows decreased confidence in patients with worse survival prospects. To further stabilize prediction variance, future research might use quantile loss optimization or heteroscedastic regression adjustments.

G. Fairness Evaluation

The subgroup-specific C-index was used to conduct a fairness analysis across six racial groups. With a standard deviation of 0.0088 and a mean C-index of 0.6838, the equity score was 0.6750. Demographic fairness is supported by the small performance gap, which shows consistent model behavior across various populations.

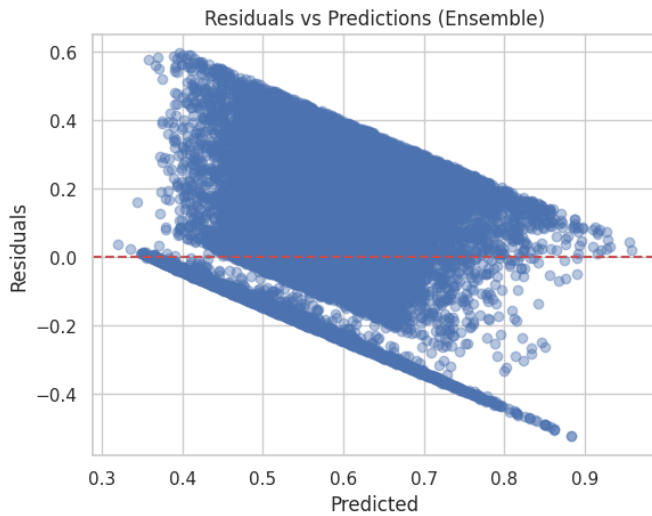


Fig. 5. Residuals vs. predictions for the ensemble model showing low variance and absence of systematic bias across predicted survival probabilities.

TABLE IV
FAIRNESS EVALUATION ACROSS RACE GROUPS

Race Group	Patient Count	C-index
More than one race	4845	0.6856
Asian	4832	0.6979
White	4831	0.6746
American Indian or Alaska Native	4790	0.6891
Native Hawaiian or Other Pacific Islander	4707	0.6777
Black or African-American	4795	0.6778
Mean / Std. Dev.		0.6838 / 0.0088
Equity Score		0.6750

V. CONCLUSION

This study presented an interpretable ensemble-learning framework for predicting event-free survival after hematopoietic cell transplantation. By combining LightGBM, two XGBoost variants, HistGradientBoosting, and Random Forest models with a Kaplan–Meier transformed target, the proposed system achieved a C-index of 0.6853 and an RMSE of 0.1930, while maintaining consistent race-wise subgroup performance with a C-index standard deviation of 0.0088. The results show that a compact ensemble can provide competitive discrimination without sacrificing transparency. Feature-importance, VIF, Kaplan–Meier subgroup, fairness, residual, and synthetic-augmentation analyses strengthened the clinical interpretation of the model and identified donor age, recipient age, conditioning intensity, GVHD prophylaxis, HLA matching, and comorbidity score as important contributors. The main limitation is the absence of direct external validation on an independent hospital or registry cohort. Future work will therefore prioritize multi-center validation, calibration analysis, prospective testing, and integration with electronic health record workflows before clinical deployment.

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